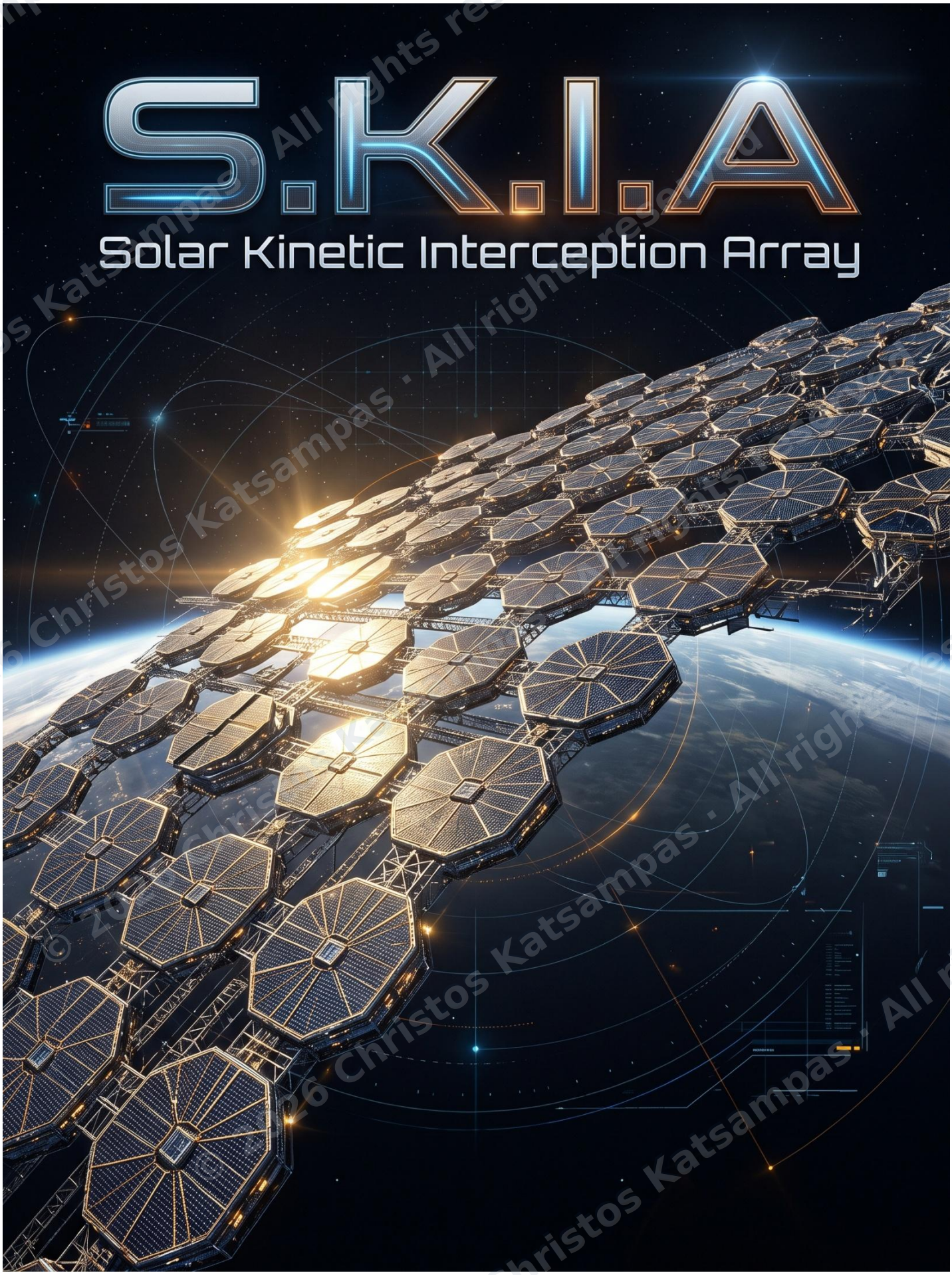



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## Abstract

SKIA (Solar Kinetic Interception Array) is two fleets built from one shared part: an octagon module 960 m wide. In geostationary orbit (GEO), 481 power stations of 10 modules each beam microwaves to ground receivers and deliver 1 TW to the grid. At the Sun–Earth L1 point, about 823,000 lighter shade modules block 0.42% of sunlight, cutting climate forcing by 1 W/m² for about 0.8 °C of long-term cooling.

The energy available is enormous. Intercepting 1% of the sunlight Earth receives captures about 1,735 TW, roughly 90 times world energy use of about 19 TW. The limits are mass to orbit, beam physics, cost and governance, not the energy budget.

Key figures from this study:

- **Module:** a square with clipped corners, 960 m wide, 0.90 km², holding 9 solar cells, 4 rim reaction wheels, 4 battery nodes and about 215,000 panels of 2 m × 2 m.
- **Power station:** 10 modules around a fixed 1 km ring, with a 0.8 km antenna and a turning beam mirror behind it; 2.08 GW to the grid, about 4,950 t. No part that carries power rotates.
- **Power fleet:** 481 stations, 1 TW, about 2.38 million t, about 17% efficient from sunlight in orbit to the grid.
- **Shade fleet:** about 823,000 modules of 2.4 t each, 1.98 million t, capped at 1 W/m² and deployed in 10 steps.
- **Receivers:** 481 graded bowl rectennas of about 39 km² each, fed by paired beams and built largely from recycled grid and power plant materials.
- **Total:** about 4.4 million t in space plus about 82 million t of mostly recycled material on the ground, roughly 43,600 reusable heavy-lift flights over about 33 years.
- **Return:** repays its carbon footprint in under 2 years and avoids about 112 Gt of $CO_2$ net over 30 years, for a build cost of $2.6–7.7 trillion (Section 13).

**Verdict:** physics plausible; engineering unproven at scale; economics and governance are the deciding unknowns. This is a concept study written from published research and first-order calculations, not a peer-reviewed paper.

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## 1. Introduction and science need

Humanity faces two linked problems: a warming climate and a fossil-dominated energy supply. A system that both cools the planet and supplies clean power addresses each directly.

**Climate forcing.** The IPCC's Sixth Assessment Report puts human-caused effective radiative forcing at 2.72 W/m² in 2019 relative to 1750 (IPCC AR6 summary). Doubling $CO_2$ from pre-industrial levels adds about 3.71 W/m² (radiative forcing overview). A sunshade reduces incoming sunlight to cancel part of that forcing.

**Energy demand.** World total energy supply passed 600 EJ in 2025, about 19 TW of continuous power, with fossil fuels still near 86% of it (Energy Institute Statistical Review 2026, via DieselNet; Connaissance des Énergies).

**Why combine them.** Both systems need the same enabling base: cheap heavy launch, thin-film structures, autonomous assembly and station-keeping. Sharing that base lowers the cost of each. They also hedge each other: the power stations cut emissions at the source, while the shade buys time against warming already locked in.

**Important limit.** A shade treats the symptom, not the cause. It does nothing about ocean acidification from $CO_2$, and if it failed suddenly, the masked warming would return within years. Roger Angel, who authored the leading sunshade paper, concluded that renewable energy remains the only permanent solution (ScienceDaily, 2006). That is why this design pairs the shade with power generation.

## 2. Physics

SKIA works because it blocks a small slice of sunlight far from Earth and generates power close to Earth. Each part of that sentence comes from a physical constraint below.

### 2.1 Sunlight is the energy source for both fleets

Sunlight arrives at 1,361 W/m² near Earth (the solar constant). Roughly half is visible light, about 45% near-infrared and a few percent ultraviolet. Light and heat are the same energy here: every wavelength warms Earth once absorbed, and every wavelength the cells convert counts as power. So a shade module cools Earth by intercepting light, and a power module produces electricity from the same light.

### 2.2 What actually causes warming

Earth absorbs about 238 W/m² averaged over its surface (1,361 ÷ 4 × (1 − 0.30 albedo)) and must radiate the same back as thermal infrared. Greenhouse gases slow that outgoing radiation, creating the forcing in Section 1. Cancelling 3.71 W/m² of forcing therefore means removing only about 1.6% of absorbed sunlight.

### 2.3 How much shading is safe

SKIA caps shading at 1 W/m², about 0.42% of absorbed sunlight. That is well under the 2.72 W/m² of forcing humans have already added, so it trims warming without overshooting into cooling below pre-industrial levels. It is deployed in steps of 0.1 W/m², each measured before the next, and every module can turn edge-on to reduce shading gradually. Section 12.1 sets out the full safety rules.

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### 2.4 Where to put the shade: L1

The Sun–Earth L1 point lies about 1.5 million km sunward, where a shade stays between the Sun and Earth all year. Three effects shape the design:

- **Penumbra.** From L1, Earth's disc is smaller than the Sun's, so a shade casts a diffuse partial shadow, not a sharp one. About 83% of a shade's blocking reaches Earth at 1.5 million km; this study assumes 72% after reflectivity and transmission losses.
- **Radiation pressure.** Sunlight pushes the shade outward. A SKIA shade module at 2.66 g/m² absorbs sunlight and accelerates at about 1.7 mm/s², so the equilibrium point shifts sunward of L1, toward Angel's proposed distance of about 1.85 million km (Angel 2006, PNAS).
- **Instability.** L1 is unstable; a shade drifts away within months unless it steers. Angel's flyers use tilting reflective panels as tiny solar sails for control.

### 2.5 Where to put the power stations: GEO, not L1

A microwave beam spreads with distance (diffraction). At 5.8 GHz from L1, a 1 km antenna makes a ground spot about 190 km wide; you would need a 20 km antenna for a 9.5 km spot. From GEO (35,786 km), a 1 km antenna gives about a 4.5 km spot. GEO stations also stay fixed over one ground site. So the shade and the power stations are separate fleets.

### 2.6 Thermal balance of the arrays

A panel converting 30% of sunlight still absorbs about 860 W/m² it must radiate from both faces. With 0.85 emissivity, it settles near 34 °C, which is acceptable for modern cells. Waste heat on the ground is negligible: using 1 TW adds only 0.002 W/m² globally, about 1/1,800 of doubled-$CO_2$ forcing.

## 3. Mathematics

Every headline number in this study follows from the six equations below, using $S_0$ = 1,361 W/m², albedo $\alpha$ = 0.30, Earth radius R = 6,371 km and Earth's cross-section $\pi R^2$ = 127.5 million km².

### 3.1 Required shading fraction

The fraction f of sunlight to block equals the forcing to cancel divided by mean absorbed sunlight:

$$f = \frac{\Delta F}{\frac{S_0}{4}(1-\alpha)} = \frac{\Delta F}{238.2\ \mathrm{W\,m^{-2}}}$$

### 3.2 Shade area

With a combined efficiency $\eta$ = 0.72 (penumbra and optics):

$$A = \frac{f\,\pi R^2}{\eta}$$

### 3.3 Shade mass

Each SKIA shade module weighs about 2.41 t over 0.904 km², an effective areal density $\sigma$ of 2.66 g/m² (2 g/m² of film plus frame, vanes and electronics). The number of modules $N_L$ and the fleet mass M follow from the area:

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$$N_L = \frac{A}{0.904\ \mathrm{km^2}}, \qquad M = \sigma A = 2.66\ \mathrm{g\,m^{-2}} \times A$$

| Scenario | Forcing offset (W/m²) | Sunlight blocked | Shade area (million km²) | Shade modules | Mass (million t) |
|---|---|---|---|---|---|
| SKIA cap | 1.00 | 0.42% | 0.74 | about 823,000 | 1.98 |
| Today's forcing (only if the cap is raised) | 2.72 | 1.14% | 2.02 | about 2.24 million | 5.39 |
| Doubled $CO_2$ (only if the cap is raised) | 3.71 | 1.56% | 2.76 | about 3.05 million | 7.35 |

For comparison, Angel's 2006 design blocked 1.8% of sunlight with about 20 million tonnes of refractive film (Angel 2006). A 2022 distributed-bubble concept cut a Brazil-sized shade to about 100,000 tonnes (space sunshade overview). SKIA's 2.66 g/m² sits between those.

### 3.4 Expected cooling

Using an equilibrium climate sensitivity of about 3 °C per $CO_2$ doubling, the long-term response is λ ≈ 0.81 °C per W/m²:

$$\Delta T \approx \lambda\,\Delta F = 0.81 \times \Delta F$$

At the SKIA cap of 1 W/m², that is about 0.8 °C of eventual cooling. Larger values in the table above would apply only if a later review raised the cap.

### 3.5 Delivered power per unit of orbital array

At GEO, sunlight is about 1,389 W/m². Multiplying the stage efficiencies (Section 7), including the beam mirror, gives 17.3% end-to-end:

$$P_{\text{grid}} = 1389 \times 0.30 \times 0.80 \times 0.99 \times 0.90 \times 0.98 \times 0.85 \times 0.97 \approx 240\ \mathrm{W\,m^{-2}}$$

That is 240 MW per km² of solar panels. A SKIA station's 8.6 km² of panels delivers 2.06 GW from the main receiving field, and the rim catcher band (Section 11.13) adds about 20 MW, bringing each site to 2.08 GW. The full 1 TW fleet carries about 4,130 km² of panels.

### 3.6 Beam spot size

The central spot diameter for a transmitter aperture D_t at distance d and wavelength λ_w:

$$D_{\text{spot}} \approx \frac{2.44\,\lambda_w\, d}{D_t}$$

At 5.8 GHz (λ_w = 5.2 cm), from GEO with a 1 km aperture: about 4.5 km. From L1: about 190 km.

### 3.7 Launch count

$$N = \frac{M}{m_{\text{payload}}}$$

With 100 t per Starship V3 flight to low Earth orbit, the shade fleet at the cap needs about 19,800 flights and the 1 TW power fleet about 23,800. Transfer from low orbit to L1 or GEO uses solar-electric propulsion or the shade film itself as a solar sail, so no extra rockets are counted for that leg.

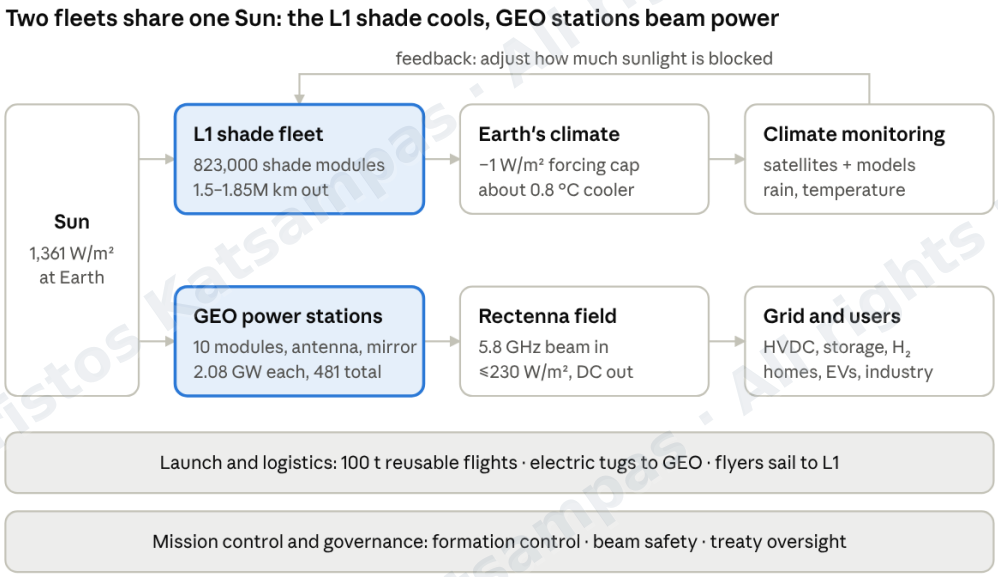
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### 3.8 Why not ship stored energy down

Physically returning batteries at about 0.3 kWh/kg would need about 29 million tonnes of returned cells per gigawatt-year delivered. That is impossible, so energy must come down as a beam.

## 4. System architecture and schematics

SKIA is two fleets built from one module design, sharing one launch chain and one control centre: a shade fleet at L1 for cooling, and a power fleet in GEO for electricity.




system architecture · 2 space fleets, 3 ground elements, 2 shared foundations

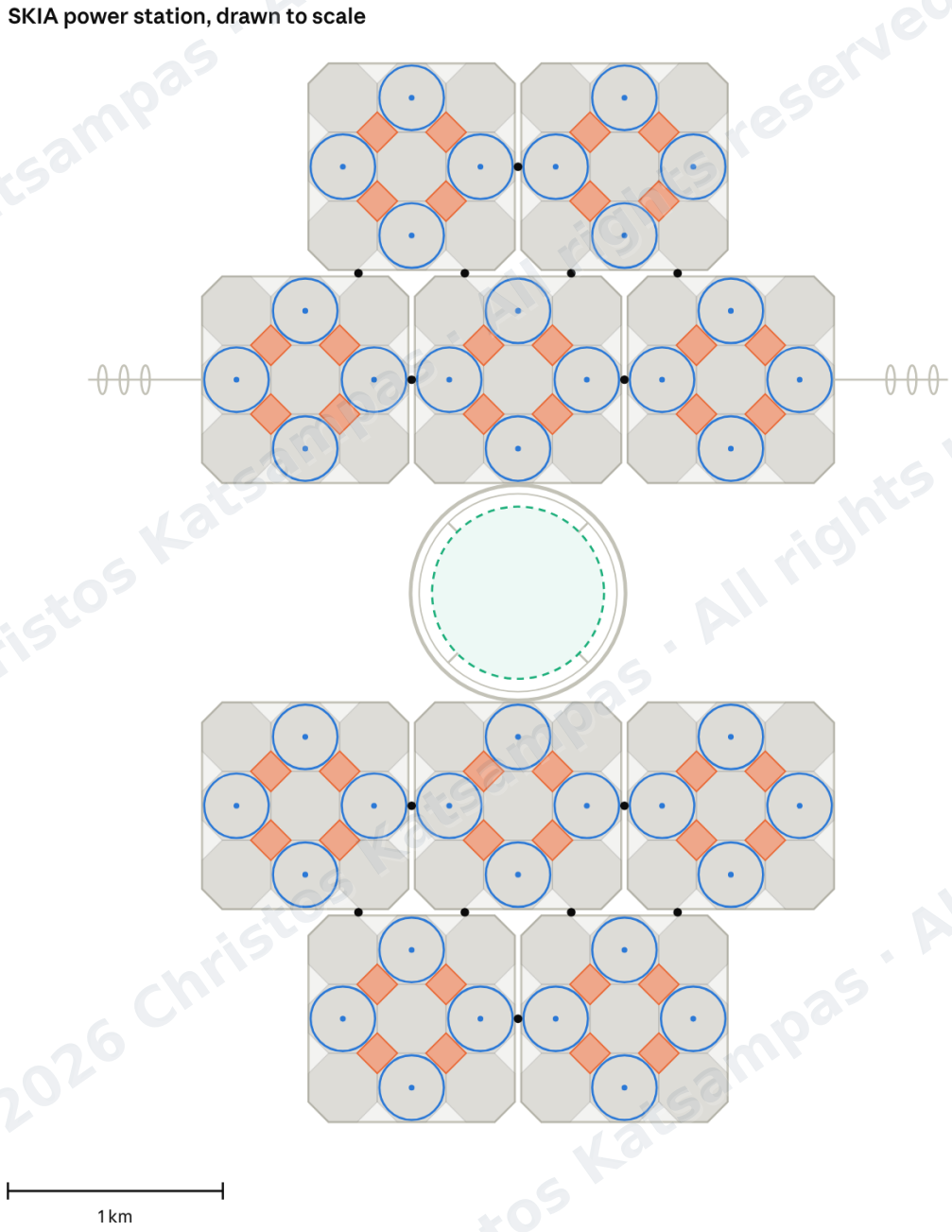
The top row is the cooling loop: monitoring measures the climate response, and mission control turns shade modules edge-on or face-on to tune how much sunlight is blocked. The bottom row is the power chain from orbit to user.

### 4.1 The SKIA module, shared by both fleets

- **Shape:** a square with clipped corners (an irregular octagon), 960 m wide. Its four long sides are 773 m and its four clipped sides 133 m; area 0.904 km², perimeter 3.6 km.
- **Cells:** 9 regular octagon cells, each 320 m across, in a 3 × 3 pattern.
- **Nodes:** 4 square nodes, 133 m across, in the gaps between cells. They hold batteries, wiring junctions and robot docks.
- **Power version:** about 215,000 solar panels of 2 m × 2 m; 4 rim reaction wheels, one in each edge-middle cell; a superconducting coil around the frame.
- **Shade version:** the same outline and cells with black film instead of panels, electrochromic control vanes at the corners and one small rim wheel.

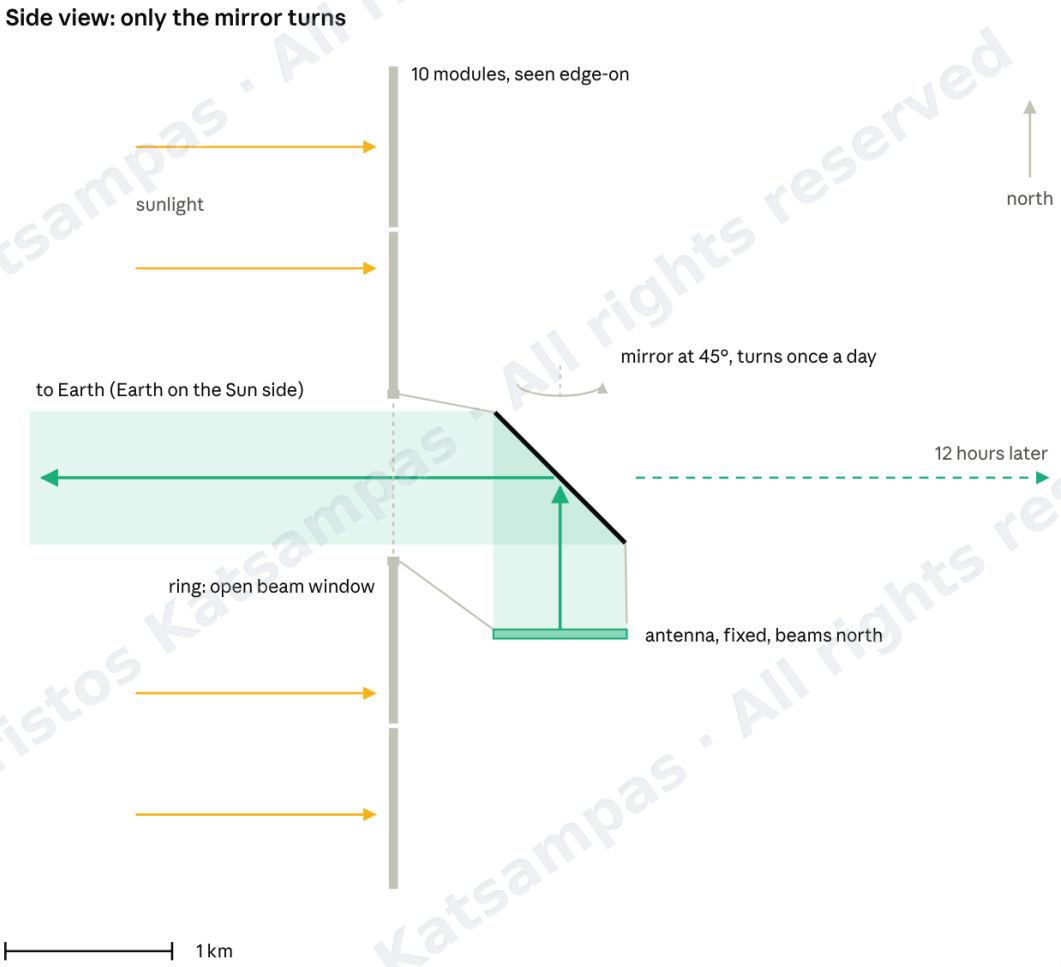
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## 4.2 The SKIA power station




SKIA power station to scale, seen from the Sun · dashed: the beam mirror about 1 km behind the ring

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SKIA power station to scale, side view from the east

The antenna beams north into the mirror, and the mirror sends the beam toward Earth. When Earth lies on the Sun side, the beam passes straight through the open ring; twelve hours later the mirror has turned half a turn and sends it the other way.

- **Layout:** 10 modules in rows of 2, 3, hub, 3 and 2, joined by tilting joints, two per shared edge.
- **Hub:** a fixed 1 km ring that frames an open beam window. Behind it, on a frame about 1 km on the shaded side, a 0.8 km antenna faces along the station's north–south axis into a 45° mesh mirror. Only the mirror turns, once a day, sweeping the beam to follow Earth; the modules and antenna never rotate.
- **Power flow:** 3.58 GW of DC flows through fixed cables to the antenna, which sends a 2.86 GW beam into the mirror; each site receives 2.08 GW at the grid (Section 11.1). Nothing that carries power rotates; when Earth is on the sunward side, the beam passes straight through the window in the ring.
- **Side booms:** part of the author's sketch, moved from the hub row to the outer edges of the rows of three so they stay out of the beam's path; their purpose is still open.

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- **Launch form:** modules ship folded with panels stacked, about 3.4 flights per module and 50 per station.

### 4.3 Ground station

- **Rectenna:** elliptical field about 6.0 × 8.3 km at 38°N (Section 11), a shallow bowl of tilted, terraced mesh rings with a catcher band on the rim (Section 11.13).
- **Centre:** pilot beacon transmitter.
- **Edge:** converter hall (DC to AC), HVDC terminal, battery bank, optional electrolyser plant.
- **Perimeter:** beam-density monitors and an aviation exclusion zone.

## 5. Hardware bill of materials

SKIA has four segments: the L1 shade modules, the GEO power stations, the ground receivers, and the launch and logistics chain. Quantities and masses for each are in Section 12; this section lists what each part is made of.

### 5.1 Shade module (Tessera-L)

One module: the SKIA outline, 960 m wide, 0.904 km², about 2.41 t. About 823,000 are needed at the 1 W/m² cap.

| Component | Function | Candidate technology |
|---|---|---|
| Shade film | Blocks sunlight | 1–1.5 µm polyimide or silicon nitride, black sunward coating, high-emissivity Earthward face |
| Frame | Keeps film flat across the 9 cells | Carbon-fibre tape-spring booms, rolled for launch |
| Control vanes | Station-keeping and attitude by sunlight pressure | Electrochromic reflectivity panels at the clipped corners |
| Rim wheel | Fine pointing | One small carbon-fibre rim in a magnetic-bearing track |
| Power | Runs electronics | Thin-film solar strip, 50–200 W, small supercapacitor bank |
| Navigation | Position within the fleet | Sun sensor, star tracker, radio ranging between modules |
| Computer and radio | Autonomous formation control and commands | Radiation-tolerant microcontroller, S-band mesh radio |
| Safe mode | Fails gracefully | Turns edge-on (no shading) on loss of contact |

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| Component | Function | Candidate technology |
|---|---|---|
| End of life | Disposal | Vanes set to drift the module into an orbit around the Sun |

Because each module fails alone and turns edge-on, a fault reduces cooling gradually instead of creating debris or sudden shading changes.

### 5.2 Power station (10 Tessera-G modules and a hub)

| Component | Function | Candidate technology |
|---|---|---|
| Solar panels | Sunlight to DC, 2 m × 2 m, swappable | Thin-film multijunction or perovskite-silicon tandem, ≥ 30% efficient |
| Cell and frame structure | Holds 9 cells and 4 nodes per module | Carbon-fibre booms and tension cables |
| Rim reaction wheels | Attitude control, 4 per module | 150 m carbon-fibre rims at 200 m/s in magnetic-bearing tracks, linear motors |
| Frame coil | Flexible wiring path and magnetic steering against Earth's field | Superconducting REBCO tape, 10,000 A loop |
| Battery nodes | Start-up power and control loads | Lithium-ion or solid-state cells, 2 MWh per module |
| Tilting joints | Link modules; follow the Sun's ±23.4° seasonal swing | Gimbaled hinges with electric actuators and dampers |
| Trunk cables | Module to antenna at 20 kV DC | Coiled superconducting cable behind a layered sunshield |
| Hub ring | Fixed frame around the open beam window; carries no power | 1 km composite ring |
| Antenna frame | Holds antenna and mirror about 1 km behind the hub, on the shaded side | Carbon-fibre truss |
| Hub antenna | DC to a 5.8 GHz beam; fixed, facing along the station's north–south axis | 0.8 km phased array, about 330 million elements in 1 m² tiles, gallium nitride amplifiers |

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| Component | Function | Candidate technology |
|---|---|---|
| Beam mirror | Turns once a day to aim the beam at Earth; carries no power | 0.8 × 1.1 km metal mesh at 45° on a magnetic bearing, with small shape actuators |
| Beam control | Locks the beam on the ground site and corrects mirror errors | Pilot-signal receivers; beam spreads harmlessly if the pilot is lost |
| Thermal | Rejects antenna waste heat | High-emissivity radiator coatings on the antenna's back face |
| Propulsion | Station-keeping | Hall-effect thrusters on krypton or xenon |
| Guidance | Pointing to 0.0005° | Star trackers, beacon tracking |
| Servicing robots | Swap panels and wheels | 0.5 t crawlers, 4 per module |

### 5.3 Ground receiving station (one per station, 2.08 GW)

| Component | Function | Candidate technology |
|---|---|---|
| Rectenna field | Microwaves to DC | Dipole antennas with Schottky rectifier diodes, about 6 × 8 km ellipse at 38°N, tilted mesh rings in a shallow bowl |
| Pilot beacon | Guides the beam | Ground transmitter at field centre |
| Inverters | DC to grid AC | Utility-scale solid-state inverters |
| Transmission | Connects to grid | HVDC link to the nearest load centre |
| Grid storage | Covers eclipse and maintenance | Batteries or pumped hydro sized for hours, not days |
| Safety | Protects people and wildlife | Peak beam density limited to about 230 W/m² (under a quarter of noon sunlight), perimeter monitors, no-fly zone |
| Electrolysers (optional) | Turns surplus into hydrogen | PEM or solid-oxide electrolyser plant on site |

### 5.4 Launch and logistics

| Element | Function | Candidate technology |
|---|---|---|
| Heavy launcher | Earth to low orbit | Fully reusable Starship V3 class, about 100 t per flight (Starship V3 overview) |

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| Element | Function | Candidate technology |
|---|---|---|
| Propellant depots | Refuel tugs in orbit | Cryogenic methane-oxygen depots |
| Orbital tugs | Low orbit to GEO | Solar-electric tugs, about 21% propellant fraction vs about 68% for chemical |
| Shade self-transfer | Low orbit to L1 | Shade modules sail themselves, about 600 days per transit (Fuglesang & de Herreros Miciano, via Warp News) |
| Factories | Mass production on Earth | Roll-to-roll film lines, chip fabs, automated tile assembly |
| Lunar supply (later phase) | Cuts launch mass | Lunar silicon and aluminium processing, mass driver launch |
| Mission control | Operates both fleets | Ground network plus international monitoring centre |
| Debris removal | Keeps GEO usable | Active removal servicers for failed modules |

## 6. Launch and deployment plan

SKIA launches power stations first and shade modules second, because power stations earn revenue that can fund the shade, and they prove the launch, assembly and control chain at smaller scale.

### 6.1 Optimisation rules

1. **Minimise mass, not launches.** Every 1 g/m² of shade film is about 744,000 t across the shade fleet at the cap. Film thickness is the single biggest lever.
2. **Launch to low orbit only.** Rockets stop at low Earth orbit. Solar-electric tugs carry power modules to GEO; shade modules sail themselves to L1. This roughly triples useful mass per rocket compared with chemical transfer.
3. **Pack flat, deploy autonomously.** Shade modules ship rolled, about 41 per flight; power modules ship folded with panels stacked. No crewed assembly.
4. **Build in modules that work alone.** Each shade module shades on arrival; each station delivers power on its own. Value starts with the first unit.
5. **One module design, two fleets.** Shared outline, cells, frame and robots mean one set of factories and one learning curve.
6. **Move manufacturing off Earth later.** Once lunar processing exists, film and structure can be made from lunar material and launched by mass driver at a fraction of the energy cost.

### 6.2 Phased launch schedule

Flight counts assume 100 t per Starship V3 class flight (Starship V3 overview).

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| Phase | Payload | Mass to orbit | Flights | Flights per year needed |
|---|---|---|---|---|
| A. Demonstrators | 1 MW beam test through a 30 m turning mirror; one power module; 100 shade modules at L1 | about 600 t | about 6 | 2–3 |
| B. First station | One 2.08 GW station (10 modules, ring, antenna and mirror) | about 4,950 t | about 50 | 10–20 |
| C. Power to 100 GW | 49 stations in total | about 243,000 t | about 2,430 | about 490 over 5 years |
| D. Shade to the cap | about 823,000 modules, 1 W/m², in 10 steps | about 1.98 million t | about 19,800 | about 2,000 over 10 years |
| E. Power to 1 TW | 432 more stations | about 2.14 million t | about 21,400 | about 1,430 over 15 years |
| F. Beyond the cap (only if a review raises it) | up to 3.05 million shade modules in total | up to 7.35 million t in total | up to 73,500 in total | Prefer lunar-made film |

For scale, SpaceX's own target cadence for 2026 is about 52 Starship launches a year (Basenor). Phases D and E need 25–40 times that, which implies many launch sites and several providers.

### 6.3 Cost drivers

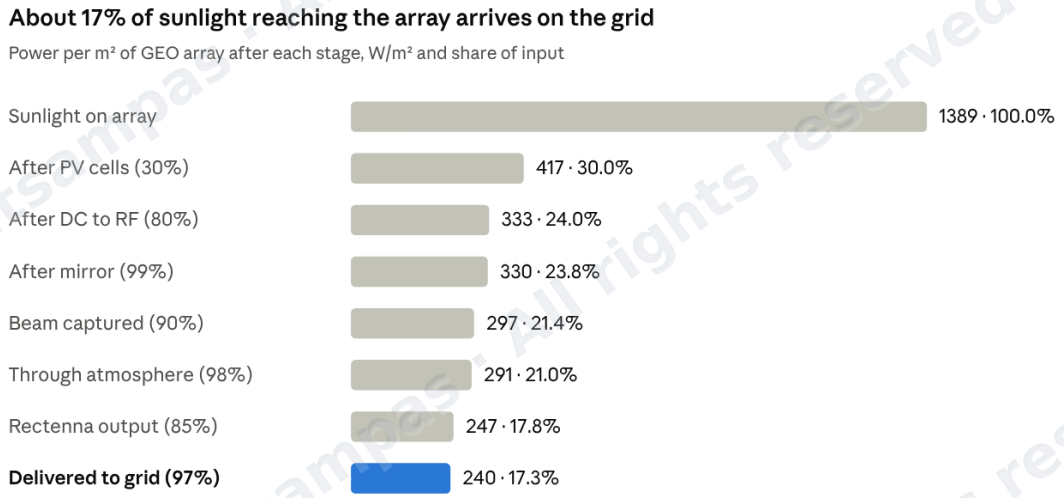
NASA's 2024 study found launch was 71–77% of total cost for a 2 GW space power system, with totals of $276–434 billion for a first-of-a-kind unit (NASA OTPS, 2024). Fully reusable launch at the low end of projected prices (about $2–10 million per flight) would cut that share sharply. At $5 million a flight, the shade fleet's 19,800 flights cost about $99 billion in launch alone, and the 1 TW power fleet's 23,800 flights about $119 billion.

## 7. Delivery outputs

The system delivers two products: steady clean electricity from the GEO fleet, and gradual global cooling from the L1 shade.

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## 7.1 Power delivered




Stage efficiencies assumed in this study (Section 3.5) · sunlight at GEO 1,389 W/m²

The biggest single loss is the solar cell itself; better cells raise every later stage proportionally. Unlike ground solar, GEO stations see the Sun about 99% of the year, with no night or clouds (NASA OTPS, 2024).

| Fleet size | Stations | Solar panel area in orbit | Power to grid | Share of world energy supply (19 TW) |
|---|---|---|---|---|
| First station | 1 | 8.6 km² | 2.08 GW | 0.01% |
| Phase C | 49 | 421 km² | 102 GW | 0.5% |
| Phase E | 481 | 4,130 km² | 1,001 GW | 5% |

The share understates the impact. Burning fuel wastes most of its energy as heat, so 1 TW of clean electricity can displace roughly 2–3 TW of fossil fuel input (approximate, depending on the use replaced).

## 7.2 How the power is used on Earth

The power arrives at fixed rectenna sites and enters the grid, so it reaches users the same way any power plant's output does:

- **Homes, industry, data centres:** direct grid supply, available day and night.
- **Electric vehicles and motors:** charged from the grid or batteries. Beaming power to individual cars, trucks or motors is not feasible; each would need a large receiving antenna and a safe, continuous beam lock.
- **Hydrogen and e-fuels:** for ships, aircraft, steel and fertiliser. 1 TW for a year at 70% electrolysis efficiency makes about 156 million tonnes of hydrogen.
- **Remote and disaster areas:** a beam can be redirected to a temporary rectenna, one of the strongest arguments for the technology.

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### 7.3 Cooling delivered

| Shade deployment | Modules at L1 | Forcing offset | Long-term cooling (approx.) |
|---|---|---|---|
| One step | about 82,000 | 0.1 W/m² | about 0.08 °C |
| Half the cap | about 411,000 | 0.5 W/m² | about 0.4 °C |
| Full cap | about 823,000 | 1.0 W/m² | about 0.8 °C |

The full response takes decades because oceans warm and cool slowly. Shading also weakens rainfall slightly, since less sunlight means less evaporation; each step is checked against regional climate models and measurements before the next.

## 8. Plausibility classification

No law of physics blocks this system; every subsystem has at least lab or small in-orbit evidence, but none has been built at anywhere near the required scale. Technology readiness levels (TRL, 1 = idea, 9 = flight-proven in operation) are this study's estimates.

### 8.1 By subsystem

| Subsystem | Evidence today | TRL (est.) | Main gap |
|---|---|---|---|
| Wireless power beaming from orbit | Caltech MAPLE beamed power in orbit and detectably to Earth in 2023 (Caltech); JAXA's OHISAMA 180 kg satellite is set to test LEO-to-ground beaming (IAC 2026) | 4–5 | Gigawatt power levels, GEO distances, beam safety |
| Lightweight space PV | Caltech tested many cell types in orbit; multijunction cells are routine on satellites | 6–7 | Cost and mass per kW at mass production |
| Deployable large structures | Solar sails such as IKAROS; Caltech's DOLCE deployment test | 4–6 | Kilometre scale, decades of life |
| Shade modules at L1 | Optical prototype built under NASA NIAC funding (space sunshade overview) | 2–3 | Station-keeping of about 823,000 modules |
| Robotic in-space assembly | Satellite servicing demonstrations | 4–5 | Fully autonomous, continuous assembly |
| Fully reusable heavy launch | Starship V3 first flew in May 2026 (Starship V3 overview) | 6–7 | Proven reuse, cadence of thousands a year |
| Rectennas | Ground demonstrations by JAXA and Japan Space Systems in 2015 and 2019 (J-spacesystems) | 6 | Land cost, siting approvals |

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| Subsystem | Evidence today | TRL (est.) | Main gap |
|---|---|---|---|
| Lunar manufacturing | Studies only | 1–2 | Everything |
| Turning beam mirror with antenna correction | Phase-correction principle proven in radar and telescope adaptive optics; never combined with a kilometre-scale mirror | 2–3 | Ground test with a 10–30 m mesh mirror, then full scale |

### 8.2 Overall rating

| Aspect | Rating | Reason |
|---|---|---|
| Physics | Plausible | Energy balance, orbits and beam physics all close |
| Engineering | Hard but credible | Each part demonstrated small; scale-up is unprecedented |
| Economics | Uncertain | NASA estimated $276–434 billion for a first 2 GW system, mostly launch (NASA OTPS); ESA's consultants found competitive prices possible by 2040 with major advances (ESA) |
| Governance | Hardest part | Who sets the global thermostat, and who controls gigawatt beams |

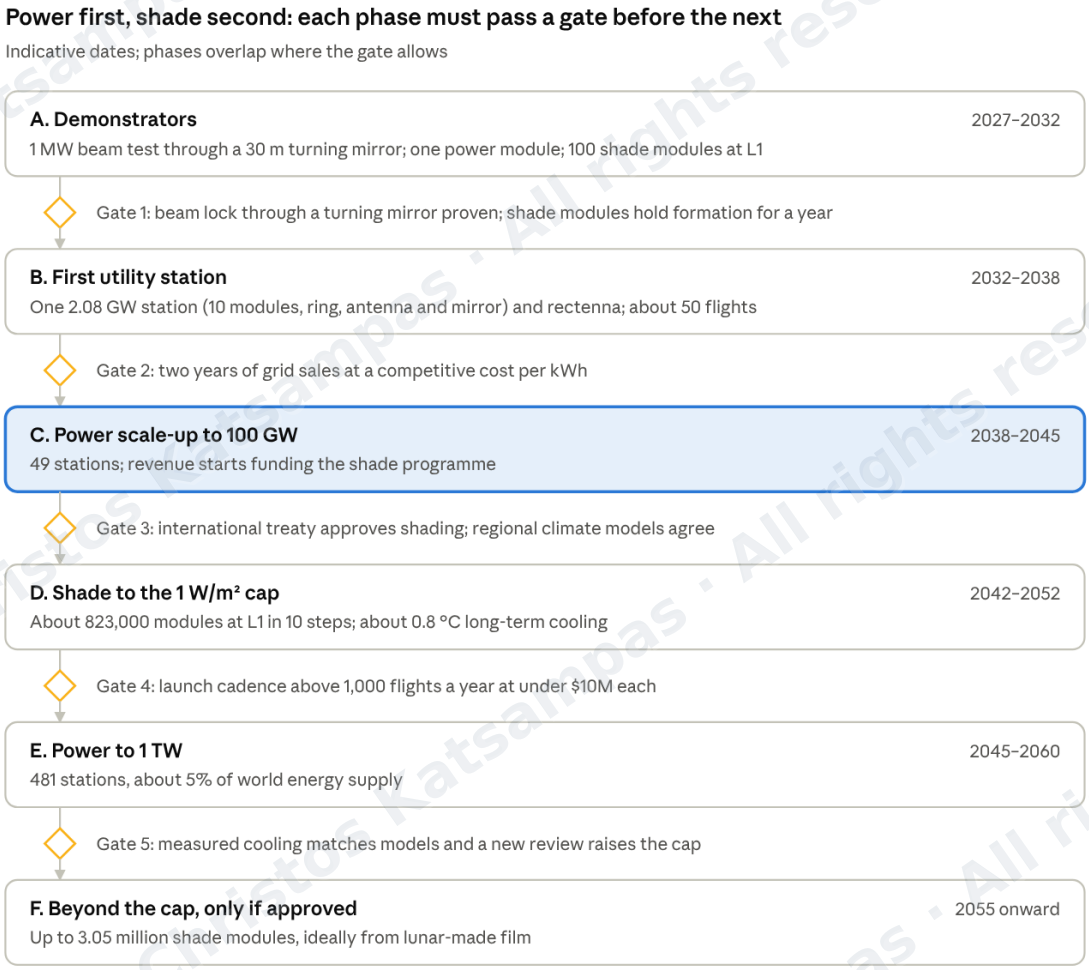
### 8.3 Key risks and mitigations

| Risk | Effect | Mitigation |
|---|---|---|
| Sudden shade failure | Rapid rebound warming (termination shock) | About 823,000 independent shade modules; no single point of failure |
| Regional rainfall change | Droughts or floods in some regions | Small phased deployment, climate monitoring, international review |
| Beam misuse or misaligned beam | Heating outside the rectenna | Low beam density by design; retrodirective lock that defocuses if the pilot signal is lost |
| Orbital debris | Damage to GEO satellites | Active debris removal, design for controlled disposal |
| Launch emissions | Soot and $CO_2$ in the upper atmosphere | Methane-oxygen fuel, later lunar supply |
| Moral hazard | Reduced pressure to cut emissions | Fund the shade only alongside the power fleet and emission cuts |
| Geopolitical conflict | Disputes over control | Treaty-based ownership, shared mission control, transparent data |

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## 9. Roadmap and conclusions

The recommended path builds power first, because it pays for itself and proves the hardware, and adds shading only after a treaty and independent climate review approve it.




development roadmap · 6 phases, 5 gates · dates indicative

Phase C is the turning point: once 100 GW is selling power, the programme can fund itself and the shade becomes a policy decision rather than a technical one.

### 9.1 Conclusions

1. **One module, two jobs.** The same 960 m SKIA module becomes a power collector in GEO or a shade at L1, so one factory line serves both fleets.
2. **A safe, tunable shade.** Capping shading at 1 W/m² and deploying it in 0.1 W/m² steps keeps the cooling small, measurable and reversible.
3. **Free energy is real in principle, not in cost.** Sunlight in space is effectively unlimited, but every kilowatt delivered needs about 2.4 kg of hardware launched and maintained.

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4. **Launch is the master variable.** About 2,000–3,000 fully reusable flights a year at a few million dollars each turn SKIA from fantasy into a hard but achievable megaproject.
5. **Governance decides the outcome.** The technology could be ready within decades; agreement on who controls the planet's sunlight may take longer.

### 9.2 Open research questions

- ☐ Lowest-mass shade film that survives decades at L1 (target ≤ 1 g/m²)
- ☐ Station-keeping algorithms for about 823,000 shade modules
- ☐ Gigawatt-class beam safety certification and ionosphere effects
- ☐ Regional rainfall response to 0.42% dimming (the SKIA cap) in high-resolution models
- ☐ Lifecycle emissions of tens of thousands of launches
- ☐ Treaty frameworks for climate intervention and power beaming

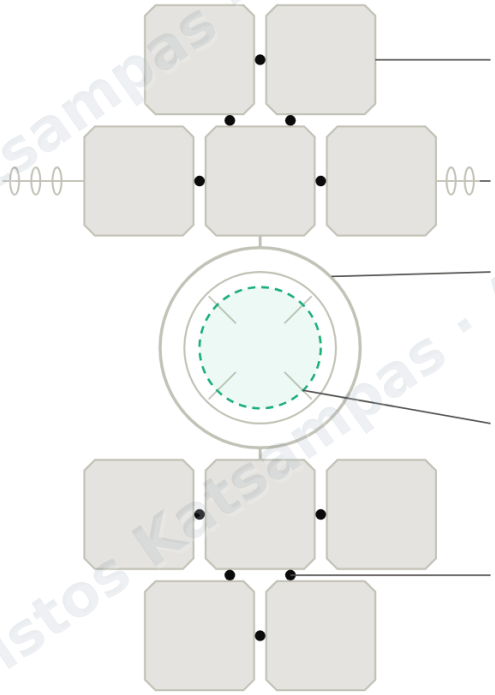
## 10. Station design: how each part works

The SKIA station comes from the author's sketches: modules around a central ring, with one beaming antenna and a turning mirror. This section explains how each part works; Section 4.2 shows it to scale and Section 12 gives its numbers.

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**SKIA station parts, labelled (not to scale)**

Redrawn from the author's sketches; below, one module in close-up



**Collector modules (10)**
clipped squares 960 m wide, 0.90 km² each

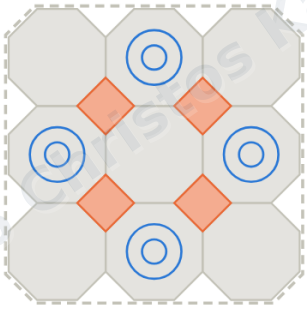
**Side booms**
purpose still open in the sketch

**Ring: fixed beam window**
the beam passes through it; nothing here rotates

**Antenna and mirror**
seen through the ring, 1 km behind it; only the mirror turns

**Tilting joints**
segway-style hinges between neighbouring modules

**One module, close-up**



Nine octagon cells with square gaps tile with no wasted space; the four battery nodes sit in the gaps, and the edge cells hold the four rim wheels.

- Solar panel cells: all nine, 2 m tiles swappable by robots
- Reaction wheels (your turbines): spin one way, the module turns the other
- Battery nodes: start-up power, control loads, wiring junctions
- Coiled cable frame: flexes as the joints tilt

SKIA station and module from the author's sketches · labelled, not to scale

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### 10.1 Module shape and size

The sketch draws each module as a square with clipped corners filled with cells. Nine octagon cells of 320 m in a 3 × 3 pattern fill exactly that shape: 960 m across, with square gaps between cells for the battery nodes. Each module carries 859,000 m² of panels, so ten modules give one 2.08 GW station. Medium-sized panels keep repairs simple: a robot swaps one 2 m panel, not a whole sheet.

### 10.2 Movement: the turbines become rim reaction wheels

A propeller does nothing in space because there is no air to push. But an electric motor spinning a heavy wheel does turn the module, in the opposite direction, because spin (angular momentum) is conserved. These are called reaction wheels; Hubble points with them, and the International Space Station uses larger versions called control moment gyroscopes.

SKIA puts one wheel in each edge cell as a 150 m rim spinning at 200 m/s in a magnetic-bearing track. A wide rim stores a lot of spin with very little mass, so each rim needs only about 45 kg of spinning material, enough to handle the extra tidal pull on the antenna and mirror held 1 km behind the panels. The wheels slowly reach top speed while fighting the steady pull of Earth's gravity across the module; the frame coil (Section 12.5) slows them back down by pushing against Earth's magnetic field, without fuel.

### 10.3 Tilting joints

The segway-style joints are gimbaled hinges with electric actuators, two per shared edge. They let modules tilt ±25° to follow the Sun's seasonal swing of ±23.4° relative to the GEO orbit, while the turning mirror handles the daily turn. A kilometre-scale structure vibrates slowly, so each joint also has a damper, so motion in one module does not ripple through the station.

### 10.4 Batteries and start-up

The four battery nodes in each module (2 MWh in total) give start-up power after deployment and run the wheels, joints and electronics. They are not sized to keep beaming through eclipses. GEO eclipses happen only near the equinoxes, last up to 72 minutes, and fall around local midnight when demand is lowest; ground storage at each rectenna covers them.

### 10.5 Wiring weight and the coil

A coiled cable, like a phone cord, is the right answer for flex: it stretches and bends as joints tilt without breaking. The weight problem is solved by voltage and material. For the same 2% loss, cable mass falls with the square of voltage. Sending one module's 358 MW about 2 km to the hub in aluminium needs:

| Line voltage | Conductor cross-section | Cable mass per module |
|---|---|---|
| 1 kV | 2.0 m² | about 21,600 t |
| 10 kV | 200 cm² | about 216 t |
| 100 kV | 2.0 cm² | about 2.2 t |

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Very high voltages risk arcing in space, so SKIA uses superconducting tape at 20 kV instead. With no resistance, the cable for one module weighs about 5 t, kept below about 50 K by a layered sunshield that faces deep space.

### 10.6 The beamer

The beam leaves the antenna at the speed of light and goes straight to Earth by way of the mirror; it does not need to circle the hub or gain speed. The antenna is fixed and faces along the station's north–south axis into a 45° mesh mirror, which turns once a day to keep the reflected beam on Earth. The mirror carries no power, so a jam only mis-aims the beam until it is fixed, and paired beams (Section 11.9) cover the gap on the ground.

The antenna's pilot-signal lock also corrects small bumps in the mirror, so the mirror needs to be smooth over each 4 cm patch, not perfect across its whole width. What sets accuracy is the antenna's width: about 0.8 km for the 6.0 km receiver in Section 11. The ring stays as a fixed frame around an open window: when Earth is on the sunward side, the beam passes straight through it.

## 11. Ground receiver design

Each SKIA station needs one rectenna. At Athens' latitude (38°N) it is an ellipse about 6.0 × 8.3 km, roughly 39 km² of land with about 58 billion antenna elements, delivering 2.08 GW to the grid. The beam is one combined beam aimed by a ground beacon, so a wider antenna makes it more accurate, not less.

### 11.1 Sizing from the station's beam

The hub antenna sends a 2.86 GW beam, and the mirror reflects 99% of it toward Earth. 98% of that, 2.78 GW, gets through the atmosphere. The rectenna catches 90%, its rectifiers turn 85% into DC, and inverters pass 97% to the grid: 2.06 GW, plus about 20 MW from the rim catcher band (Section 11.13), for 2.08 GW.

The beam's power density falls off from its centre in a bell-shaped (Gaussian) profile. With peak density $I_0$ at the centre and beam radius w:

$$I(r) = I_0\, e^{-2r^2/w^2}, \qquad P_{\text{beam}} = \frac{\pi w^2 I_0}{2}$$

Capping the centre at $I_0$ = 230 W/m² gives w = 2.77 km. Catching 90% of the beam needs a rectenna radius of:

$$R_{90} = w\sqrt{\tfrac{\ln 10}{2}} = 2.98\ \mathrm{km}$$

So the receiving field is a 6.0 km circle when the satellite is straight overhead. Density is 230 W/m² at the centre, 23 W/m² at the edge, and about 80 W/m² on average.

### 11.2 Latitude stretches the field

A GEO station sits over the equator, so farther from the equator the beam arrives at a lower angle and its footprint stretches north–south.

| Site latitude | Elevation of station | Field size (km) |
|---|---|---|
| 0° (equator) | 90° | 6.0 × 6.0 |

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| Site latitude | Elevation of station | Field size (km) |
|---|---|---|
| 20° | 67° | 6.0 × 6.5 |
| 38° (Athens) | 46° | 6.0 × 8.3 |
| 50° (London, Kyiv) | 33° | 6.0 × 11.1 |
| 60° (Oslo, Helsinki) | 22° | 6.0 × 16.0 |

Low-latitude sites are much cheaper in land. Northern countries can receive power at a southern rectenna and import it over HVDC lines.

### 11.3 Rectenna construction

The field is built from identical blocks, the same modular logic as the SKIA modules in orbit:

- **Element:** a half-wave dipole antenna tuned to 5.8 GHz (wavelength 5.2 cm), spaced about 2.6 cm apart, about 1,500 per m².
- **Rectifier:** one Schottky diode per element (gallium arsenide or gallium nitride) turns microwaves into DC, with a small filter that stops the diode re-radiating.
- **Panel:** elements printed on a mesh sheet that lets most light and rain through, mounted on posts in tilted, terraced rings across a shallow bowl (Section 11.13).
- **Block:** 100 m × 100 m of panels wired in series and parallel. A centre block produces about 2 MW; an edge block about 0.2 MW. Failed blocks are swapped like modules in orbit.
- **Collection:** DC cables from each block to a central converter hall, then inverters and an HVDC terminal to the grid.
- **Storage:** 2.5 GWh of storage (Section 11.10) covers equinox eclipses around local midnight and maintenance breaks.

### 11.4 Accuracy: one beam, not many

The hub antenna's roughly 500,000 tiles do not send separate beams. Each tile times its microwaves so that all the waves add up at one point on the ground, forming one beam. The ground beacon at the rectenna centre sends a pilot signal up; every tile reads its arrival timing and transmits back along the same path (retrodirective control). The pilot signal reaches the tiles by way of the mirror, so the same lock also corrects small errors in the mirror's shape and pointing.

To keep the beam within 5% of the field width, pointing must hold to about 8 microradians, or 0.0005°. Japan's OHISAMA mission aims for an error under 0.001° from low orbit (HWBusters). If the pilot signal stops, the tiles lose their timing and the beam spreads out harmlessly instead of wandering.

### 11.5 Safety zone

The centre density of 230 W/m² is below a quarter of noon sunlight, but it is above general public microwave exposure limits, which are roughly 10 W/m² at these frequencies (approximate; national rules vary). So the field is fenced and closed to the public, aircraft are routed around it, and density falls to 10 W/m² about 3.5 km from the centre, roughly 0.5 km beyond the field edge. That ring is a buffer of farmland or solar farm with restricted access.

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### 11.6 Scale-up

At about 39 km² per station, the 481 rectennas of the 1 TW fleet cover roughly 18,800 km², a little less than the area of Wales. Spread across the world's sunbelt, deserts and offshore platforms, that is a large but manageable footprint; ground solar of the same average output would need several times more land because it stops at night and under clouds.

### 11.7 What SKIA demands of its receivers

SKIA's central-hub design puts constraints on the ground that receivers built for other designs do not face. Nine of them shape the SKIA receiver; each has an answer below.

| SKIA feature | What it means on the ground | Problem it creates | SKIA receiver answer |
|---|---|---|---|
| One 0.8 km hub antenna per station | One 2.86 GW beam lands on one site | Splitting power among smaller sites does not shrink each field, because spot size is set by antenna width, not power | One 39 km² rectenna per station, shared with a neighbour (paired beams, 11.9) |
| Hub is the only beamer | A hub fault cuts 2.08 GW at once | Larger than the biggest sudden loss many grids are built to survive | Paired beams halve the loss to about 1.04 GW (11.9) |
| Bell-shaped beam, 230 W/m² centre to 23 W/m² edge | Each antenna element gets 0.15 W at the centre but 0.015 W at the edge | Diodes lose efficiency at low power; today's 5.8 GHz rectennas reach about 70%, below SKIA's 85% target | Graded rectenna: group edge elements before each diode (11.8) |
| Station batteries cover start-up only | GEO eclipses (up to 72 min, near equinoxes) stop the beam | Each site goes dark for up to 72 minutes around local midnight | 2.5 GWh storage per site plus the SKIA ring grid (11.10) |
| Fixed 0.8 km antenna | Field size fixed at 6.0 km across overhead | Fields stretch to 11 km at 50° and 16 km at 60° latitude | Sites below about 50°; northern countries import over HVDC |
| 481 stations along the GEO ring | Each needs its own site within view, about every 0.75° of longitude | Beams and their faint side-rings must not overlap other sites | Site spacing plan and side-ring monitoring at every fence |
| 5.8 GHz beam | Shares the band with WiFi; diodes re-radiate at multiples of 5.8 GHz | Interference with local radio users | Harmonic filters in every element; a dedicated frequency allocation for power beaming |

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| SKIA feature | What it means on the ground | Problem it creates | SKIA receiver answer |
|---|---|---|---|
| Station drifts slightly within its orbital slot | The beam source moves by kilometres | The spot would wander off the field | Retrodirective pilot lock re-aims continuously (11.4) |
| Ground side is heavy | About 170,000 t per rectenna, about 82 million t for the fleet | Twenty times the mass of everything in space | Build it largely from recycled grid and plant materials (11.11) |

### 11.8 The graded rectenna

A rectifier diode works best when it receives at least about 0.1 W; below that its efficiency falls. In SKIA's beam, a single antenna element gets 0.15 W at the centre but only 0.015 W at the edge. The SKIA receiver therefore changes how elements are wired as it moves outward: one diode per element in the centre, then 2 to 4 elements combined before each diode, then up to 10 at the edge. Every diode then sees at least 0.1 W.

| Rectenna at 38°N | One diode per element | Graded and tilted (SKIA bowl) |
|---|---|---|
| Diodes per site | about 58 billion | about 19.5 billion |
| Gallium for 481 sites (gallium arsenide diodes) | about 715 t | about 240 t |

The graded design is how SKIA aims for its 85% rectifier target, against about 70% in most 5.8 GHz rectennas built so far (IAF archive). Together with the tilted rings of Section 11.13, it also cuts gallium demand by about two thirds. That matters: the world produced about 900 t of primary gallium in 2025, 99% of it in China (USGS 2026). Spread over 30 years, SKIA's 240 t is under 1% of world output a year; gallium nitride or silicon diodes are fallbacks if supply tightens.

### 11.9 Paired beams

The hub antenna can form two beams at once. In SKIA each station sends half its power to each of two rectennas, and each rectenna receives two half-beams from two neighbouring stations. The density on the ground is unchanged, but if one hub or its mirror fails, each of its two sites loses only half its supply, about 1.04 GW, and storage covers that half while the grid adjusts. The cost is a small loss from splitting the beam, still to be measured.

### 11.10 Storage and the SKIA ring grid

A GEO eclipse lasts up to 72 minutes and happens near the equinoxes around local midnight at the station's longitude. Covering it locally takes 2.08 GW × 72 minutes, about 2.5 GWh of storage per rectenna.

The SKIA ring grid shrinks that need. Stations about 18° of longitude apart never eclipse at the same time, so rectennas that far apart (about 1,600 km at 38°N) can cover each other over HVDC links. With

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the ring grid built, local storage only has to bridge the switchover; second-life electric vehicle batteries or sodium-ion cells can supply it without new lithium mining.

### 11.11 Materials budget and recycled sources

Each rectenna weighs about 170,000 t, mostly steel posts that hold the mesh up. The ground side of SKIA is about 82 million t, twenty times the mass in space, and almost all of it is common metal that retired infrastructure already contains. These are first-order estimates.

| Item | Per rectenna | 481 sites | Main recycled source |
|---|---|---|---|
| Steel posts and frames (4 kg/m²) | 156,000 t | 75 million t | Retired pylons, power plant structures, scrap steel |
| Mesh substrate (100 g/m²) | 2,800 t | 1.35 million t | Glass and polymer from retired solar panels |
| Aluminium DC collection cable | 6,000 t | 2.9 million t | Retired overhead power lines |
| Copper in converters and HVDC | 3,000 t | 1.4 million t | Transformers and generators from retired plants |
| Aluminium antenna traces (20 g/m²) | 560 t | 0.27 million t | Overhead lines, solar panel frames |
| Rectifier diodes | 19.5 billion | 9.4 trillion | Partly from electronics recycling (gallium) |
| Storage | 2.5 GWh | 1.2 TWh | Second-life electric vehicle batteries |

Retired solar panels alone are expected to reach about 78 million t of waste by 2050, with recoverable materials worth about $15 billion (IRENA and IEA-PVPS, via pv magazine). SKIA turns that waste stream into receiver material.

### 11.12 Siting at retired power plants

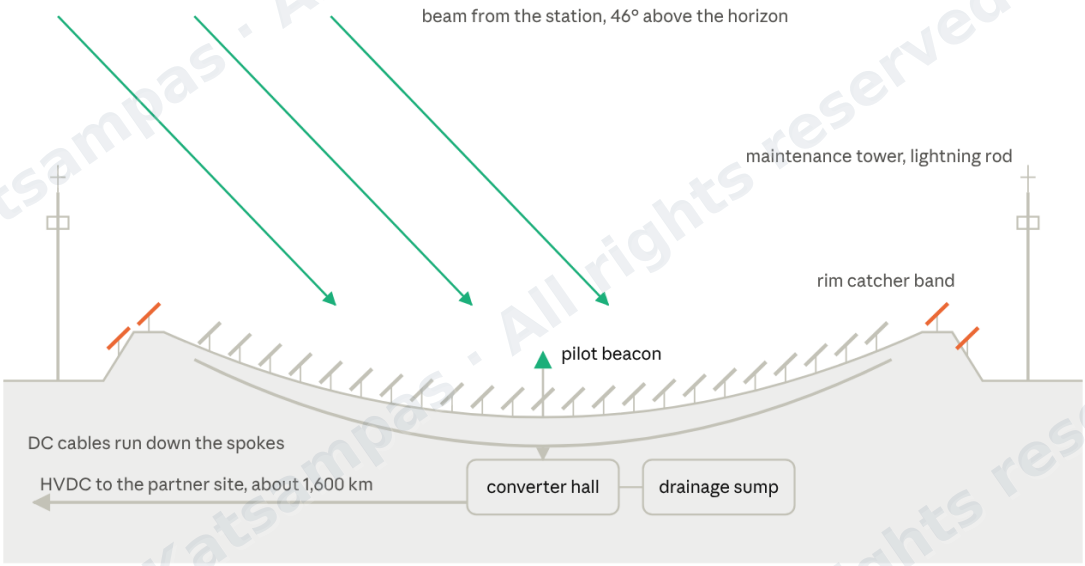
A retired coal or gas plant is a natural home for a rectenna's ground equipment. It already has a high-voltage grid connection, the right to send power into the grid, cleared industrial land and a local workforce, and getting a grid connection is often the slowest step in building any new power plant. A 39 km² field is usually larger than the plant site, so the field goes on neighbouring land or offshore platforms while the converter hall, storage and grid link take the old site. The plant itself is dismantled on site, and its steel and copper go straight into the rectenna.

### 11.13 The SKIA bowl receiver

The SKIA receiver follows the author's sketch: a shallow bowl whose inside is the main receiver, a catcher band around the rim, a pilot beacon at the centre, maintenance towers with lightning rods, and cables to a partner site.

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**SKIA bowl receiver: tilted rings face the station**




SKIA bowl receiver in cross-section, from the author's sketch · not to scale

**Shape.** A true half sphere over a 6.0 × 8.3 km field would rise about 3 km, so the bowl is shallow. Its rim is an earth berm about 20–30 m high, built from the soil dug out for the underground halls and cable trenches.

**Natural bowls.** Old open-pit mines already have this shape: wide, terraced pits, often next to retired power plants that still have grid connections. Greece's lignite basin around Ptolemaida and Kozani in Western Macedonia is one candidate as the country phases out lignite. Reusing a pit spares farmland and turns its benches into ready-made rows. Because a rim of height h shades about h metres of floor from a beam arriving at 46°, only pits or benches shallower than about 100 m are usable.

**Tilted rings.** The inside of the bowl is terraced into rings of mesh panels, each row tilted about 44° to face the station. The beam then meets every panel square on, so the mesh area falls from 39 km² to about 28 km² and the diode count to about 19.5 billion. Rows are spaced so they do not shadow each other from the beam.

**Channelling to the centre.** Power flows to the centre through cables, not by focusing the beam: DC cables run down the bowl's spokes to the converter hall under the beacon. A concave shape could act as a mirror and concentrate stray reflections at the centre; because each row faces the station, the little the mesh reflects goes back up the beam instead.

**Rim catcher band.** About 10% of the beam falls outside the main field, spread thinly all around it. A continuous band of many small, heavily grouped antennas on the rim and beyond, out to where the beam falls to 10 W/m² about 3.5 km from the centre, catches about 5.7% of the beam. That is roughly 105 MW more to the grid: about 20 MW of it offsets the beam mirror's loss and counts toward each site's 2.08 GW, and the remaining 85 MW is kept as margin, and it makes the land beyond the band safe for the public. The band costs about 14 km² of land and about 56,000 t of extra steel.

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**Towers and lightning.** Eight maintenance towers stand around the rim about 2.7 km apart, each carrying a lightning rod, a weather station and a dock for inspection drones; ground robots work under the mesh. Lightning protection for the field itself comes from bonding every steel post to a buried grounding grid.

**Weather.** Panels are rated for hail, and the open mesh lets wind through. The beam's waste heat, about 35 W/m² at the centre, helps clear snow near the middle. Robots clean dust at desert sites, and the station can turn the beam down during severe storms.

**Water.** A bowl collects rain, so runoff drains to a sealed sump beside the converter hall. The water then cools the converters, and both halls are sealed against flooding.

**Underground.** Only the mesh must stay above ground, because soil absorbs microwaves. The DC cables, converter hall, storage hall and HVDC line all go below it. A fully public-safe field is also possible by lowering the peak from 230 to 10 W/m², but it then grows to about 29 km across, more than 20 times the land.

**The partner site.** The cables leaving the bowl are the HVDC link to the partner rectenna about 1,600 km away in the SKIA ring grid (Section 11.10). The site's power also arrives from two stations at once through paired beams (Section 11.9).

### 11.14 Receiver questions still open

- ☐ Measured efficiency of a graded rectenna at 5.8 GHz across 15–150 mW per diode
- ☐ Power lost when the hub antenna splits into two beams
- ☐ A frequency allocation for power beaming and coexistence with WiFi
- ☐ Wind and snow loading on 39 km² of raised mesh
- ☐ Effects on birds and insects living under and near the mesh
- ☐ Grid codes that accept a 1.04 GW sudden loss with storage backing it

## 12. SKIA specification

SKIA is two fleets built from one module design: 4,810 power modules in GEO delivering 1 TW to the grid, and about 823,000 shade modules at L1 cutting forcing by 1 W/m². Together they weigh about 4.4 million tonnes, roughly 43,600 heavy-lift flights. All figures come from the equations in Sections 3 and 11; the mass rates are design targets, not proven values.

### 12.1 Targets and the safety rules

| Target | Value | Why this level |
|---|---|---|
| Power to grid | 1 TW | About 5% of world energy supply; displaces roughly 2–3 TW of fossil fuel input |
| Shading cap | 1.0 W/m² (0.42% of sunlight) | About 0.8 °C long-term cooling, well below the 2.72 W/m² humans have already added |
| Shading steps | 0.1 W/m² each (about 82,000 modules) | Small enough to measure the climate response before the next step |

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| Target | Value | Why this level |
|---|---|---|
| Ramp-down limit | No more than 0.1 W/m² removed per year | Prevents rebound warming if the programme stops |
| Failure mode | Each module fails alone, turning edge-on | No single fault can switch off the shade at once |
| Raising the cap | Only after Gate 5 (Section 9) | Measured cooling must match models first |

Only the L1 fleet cools Earth. A GEO station sits between the Sun and Earth for only a small part of each day near the equinoxes, so the power fleet intercepts almost no sunlight that would have reached Earth. At L1, every square metre of a module shades Earth, whether it is film, a wheel or a node.

### 12.2 Fleet counts

| Item | Quantity |
|---|---|
| Power stations (Kypseli), 10 modules each | 481 |
| Power modules (Tessera-G) | 4,810 |
| Solar panels, 2 m × 2 m | about 1.03 billion |
| Hub antennas (0.8 km) and beam mirrors (0.8 × 1.1 km), one each per station | 481 |
| Ground rectennas | 481 |
| Shade modules at L1 (Tessera-L) | about 823,000 |
| Servicing robots (Melissa) in GEO | 19,240 (4 per power module) |

The first 100 GW is 49 stations.

### 12.3 Module geometry (shared by both fleets)

| Feature | Size |
|---|---|
| Module outline | Square with clipped corners, 960 m wide; long sides 773 m, clipped sides 133 m |
| Module area | 0.904 km²; perimeter 3.6 km |
| Cells | 9 regular octagons in a 3 × 3 pattern, each 320 m across and 84,800 m² |
| Square nodes | 4, each 133 m × 133 m, in the gaps between cells |
| Panel | 2 m × 2 m square, 4 m², about 1 kg |
| Panels per power module | about 215,000, covering 95% of the module (859,000 m²) |

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### 12.4 Power module (Tessera-G) mass budget

Each power module collects 358 MW of DC power, which becomes 208 MW at the grid after beaming. It weighs about 339 t, about 3.4 flights.

| Component | Specification | Material | Mass (t) |
|---|---|---|---|
| Solar panels | 859,000 m² at 30% efficiency, 250 g/m² | Thin-film multijunction or perovskite-silicon tandem | 215 |
| Structure | Tape-spring booms and cell frames, 50 g/m² | Carbon-fibre composite | 45 |
| Internal wiring | Panel strings to cell buses to nodes | Aluminium, medium voltage | 16 |
| Battery nodes (4) | 2 MWh total for start-up and control loads | Lithium-ion or solid-state cells | 8 |
| Rim reaction wheels (4) | 150 m radius rims at 200 m/s in magnetic-bearing tracks | Carbon-fibre rim, linear motors | 8 |
| Avionics, thermal, thrusters | Computers, radios, star trackers, Hall thrusters | Mixed | 5 |
| Trunk cable to hub | 20 kV DC, 17,900 A, 54 superconducting tapes per pole, 2 km | REBCO tape with radiative sunshield | 5 |
| Joint share | Half of each hinge shared with neighbours | Titanium and composite, with dampers | 4 |
| Servicing robots (4) | 0.5 t crawlers that swap panels | Mixed | 2 |
| Frame coil | 10,000 A loop, 30 superconducting tapes, 3.6 km | REBCO tape | 1 |
| Margin | 10% | — | 31 |
| **Total** | | | **339** |

The rim wheels are cheap because their size works in their favour. Steadying a module against the tidal pull of Earth's gravity needs about 1.5 million N·m·s of stored spin on its own, and about 5.5 million once the station's antenna and mirror are held 1 km behind the panels. Four rims 150 m in radius store that with about 180 kg of spinning mass in total; the rest of the 8 t is track and motors. At 200 m/s the rim's material stress is 64 MPa, a small fraction of carbon fibre's strength.

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**Is 250 g/m² of panels realistic?**

On paper yes, with today's best thin solar cells; in flown hardware not yet, as no space solar array built so far comes within about ten times of it. SKIA keeps 250 g/m² as its design target and carries a conservative case of 500 g/m² alongside it.

At 30% efficiency in GEO sunlight, 250 g/m² means each kilogram of panel delivers about 1,670 W of DC power. Real hardware for comparison:

| Technology | Power per kg | What is counted | Source |
|---|---|---|---|
| Roll-out solar array (ROSA), flown on the International Space Station | 100–120 W | Whole array, including structure | Redwire, via SatNow |
| NASA ultralight thin-film array study (2007) | 362 W near term, 500 W projected | Whole wing; blankets alone weigh 600 g/m² | NASA NTRS |
| Thin gallium arsenide cells grown and peeled off a reusable wafer (MicroLink) | over 3,000 W | Bare cells only | Semiconductor Today |
| Lab perovskite cells on 1–2 µm plastic film | 23,000–29,000 W | Tiny bare cells, no protective layer | Review, arXiv |

The thin gallium arsenide cells are the realistic base. At 3,000 W per kg they weigh about 136 g/m², which leaves room inside 250 g/m² for the rest of the panel:

| Panel layer | Mass (g/m²) |
|---|---|
| Thin gallium arsenide cells | 136 |
| Radiation-resistant coating instead of cover glass | 20 |
| Cell wiring and connectors | 30 |
| Polyimide backing film, about 12 µm | 17 |
| Panel edge frame and mounting clips | 40 |
| **Total** | **243** |

Perovskite cells are lighter still, but Caltech's in-orbit test found them unreliable in space, while gallium arsenide cells performed consistently (Popular Science). They stay a later upgrade, not the baseline.

The thin coating lets more radiation through than cover glass, so cells will wear faster over 30 years in GEO. SKIA's design already allows for that: the servicing robots swap 2 m panels as they wear, and the replacement mass belongs in operations and maintenance (Section 13.6).

**If panels come out heavier**, more than the panels changes, because launches, cost and carbon all follow mass:

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| Panel mass | Power module | Station | Total in space | Flights | Hardware per kW |
|---|---|---|---|---|---|
| 250 g/m² (target) | 340 t | 4,950 t | 4.36 million t | about 43,600 | 2.4 kg |
| 375 g/m² | 458 t | 6,130 t | 4.93 million t | about 49,300 | 2.9 kg |
| 500 g/m² (conservative) | 576 t | 7,310 t | 5.50 million t | about 55,000 | 3.5 kg |
| 1,000 g/m² | 1,048 t | 12,040 t | 7.77 million t | about 77,700 | 5.8 kg |

Even at 1,000 g/m², a SKIA power module still delivers about 340 W per kg, roughly three times ROSA's figure. Heavier panels make SKIA slower and costlier but do not break it; Section 13.6 shows the cost effect.

One optional twist could save more mass. Thin mirror film, about 10 g/m², angled along the cell edges could focus 1.5 times more sunlight onto each cell, so a third less cell area is needed. That saves about 68 t per module, but the cells run at about 70 °C instead of 34 °C and lose some efficiency, so overall it is about 14% lighter per kW delivered. It is worth testing, not adopting yet.

### 12.5 The frame coil does three jobs

The coil around each module frame is flexible wiring, a current loop and a steering system. A 10,000 A loop around 0.904 km² makes a magnetic field strong enough to push against Earth's magnetic field at GEO, giving up to about 980 N·m of torque. That is still more than the roughly 800 N·m of gravity torque per module, including the share from the antenna and mirror, so the coil can slow the rim wheels without using fuel.

As energy storage the coil is small, about 0.1 kWh per module, so storage stays in the battery nodes and on the ground.

### 12.6 Station hub (one per station)

| Component | Specification | Mass (t) |
|---|---|---|
| Beaming antenna | 0.8 km diameter, 503,000 m², about 330 million elements in 1 m² tiles; fixed, facing the mirror | 1,005 |
| Beam mirror | 0.8 × 1.1 km elliptical metal mesh at 45°, turning once a day on a magnetic bearing, with shape actuators | 250 (est.) |
| Antenna frame | Truss holding antenna and mirror about 1 km behind the hub, on the shaded side | 150 (est.) |
| Hub ring | Fixed 1 km ring framing the open beam window | 100 (est.) |
| Hub structure and control | Spokes, computers, beacon receivers | 50 (est.) |
| **Hub total** | | **about 1,555** |

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The antenna turns 3.58 GW of DC into a 2.86 GW beam and sheds the rest as heat, 1.4 kW/m², settling near 75 °C, within the range of gallium nitride amplifiers; sitting in the panels' shadow helps it stay cool. The mirror reflects 99% of the beam toward Earth. A full station of 10 modules plus hub weighs about 4,950 t (50 flights), or 2.4 kg per kW delivered.

### 12.7 Connectors between modules

| Item | Specification |
|---|---|
| Tilting joints | Two hinge points per shared edge, each with an electric actuator and a damper |
| Tilt range | ±25°, to follow the Sun's seasonal swing of ±23.4° relative to GEO |
| Trunk cables | Coiled superconducting cable across each joint, so it flexes without strain |
| Cable cooling | Passive: the cable sits behind a layered sunshield facing deep space, held below about 50 K |
| Mechanical link | Joints carry tension only; each module's own wheels hold its attitude |

### 12.8 Shade module (Tessera-L)

The L1 version keeps the same outline and cells so both fleets share factories and robots, but swaps solar panels for black film.

| Component | Specification | Mass (t) |
|---|---|---|
| Shade film | 0.904 km², 2 g/m², black sunward face, high-emissivity Earthward face | 1.81 |
| Frame and booms | Lighter composite frame | 0.40 |
| Control vanes | Electrochromic panels at the clipped corners | 0.15 |
| Electronics and one small rim wheel | Computer, radio, sun sensor, power strip | 0.05 |
| **Total** | | **about 2.41** |

At the 1 W/m² cap: about 823,000 modules, 1.98 million t, about 19,800 flights at 41 modules per flight. Each 0.1 W/m² step is about 82,000 modules.

### 12.9 Totals

| Fleet | Units | Mass | Flights at 100 t |
|---|---|---|---|
| Power (1 TW) | 481 stations | 2.38 million t | about 23,800 |
| Shade (1 W/m² cap) | 823,000 modules | 1.98 million t | about 19,800 |

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| Fleet | Units | Mass | Flights at 100 t |
|---|---|---|---|
| **SKIA total** | | **4.36 million t** | **about 43,600** |

Ground receivers add about 82 million t, mostly recycled steel (Section 11.11).

### 12.10 A future puzzle piece: harvesting at L1

The shade fleet at the cap intercepts about 1,000 TW of sunlight, 50 times world energy use, and today it is thrown away as heat. A microwave beam from L1 spreads too much (Section 2.5), but a laser beam does not: at about 1 micrometre wavelength, a 10 m laser aperture at L1 makes a spot only about 390 m wide at Earth's distance. Adding thin solar strips and lasers to some shade modules could send power to the GEO stations, which then beam it down. This is the next idea to size, not part of the baseline above.

### 12.11 Numbers to verify first

- ☐ Panel mass of 250 g/m² including wiring and mounts: analysed in Section 12.4; achievable with today's thin gallium arsenide cells but unflown at array level, so the phase A power module must measure it
- ☐ Superconducting tape staying below 50 K behind a warm panel
- ☐ Rim wheel bearings and tracks over 30 years
- ☐ Antenna correction of a 1 km mesh mirror, tested first on the ground with a 10–30 m mirror
- ☐ Shade film at 2 g/m² surviving decades of sunlight and particles
- ☐ Radio frequency coordination for 481 stations in the GEO ring

## 13. Costs, risks and environmental return

Building SKIA would take about 33 years, $2.6–7.7 trillion, roughly 510,000–680,000 workers and about 86 million t of material, most of it recycled steel on the ground. In return, the power fleet repays its carbon footprint in under 2 years and its build energy in about 2 months, then avoids about 112 Gt of $CO_2$ net over 30 years, while the shade adds about 0.8 °C of cooling for as long as it is maintained. Every figure here is a first-order estimate with wide uncertainty.

### 13.1 Time

The roadmap in Section 9 runs from demonstrators in 2027 to the full 1 TW power fleet around 2060, with the shade reaching its cap around 2052. Power revenue starts in the late 2030s, about a decade before the shade begins.

### 13.2 Materials

| Segment | Mass | Scarce or critical materials |
|---|---|---|
| Power fleet in GEO | 2.38 million t | Gallium (amplifiers), indium and germanium or lead and iodine (solar cells), silver and rare earths such as yttrium (superconducting tape), xenon or krypton (thrusters) |

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| Segment | Mass | Scarce or critical materials |
|---|---|---|
| Shade fleet at L1 | 1.98 million t | Mostly abundant: polymer or silicon nitride film, carbon fibre |
| Ground receivers | about 82 million t | Mostly steel, aluminium and copper; about 240 t of gallium for diodes |
| **Total** | **about 86 million t** | |

The ground side is the heaviest but least exotic part, which is why the recycling plan in Section 11.11 targets it. The space side is light but needs scarce elements in small amounts per unit across very large numbers of units, so supply chains for gallium and superconducting tape must grow early.

### 13.3 Energy to build, and energy payback

| Item | Energy |
|---|---|
| Launch propellant and its production, about 43,600 flights at about 60 TJ each | about 2.6 EJ |
| Space hardware, at an assumed 0.5 GJ per kg | about 2.2 EJ |
| Ground receivers, reduced by recycled metals | about 0.5 EJ |
| **Total to build** | **about 5.3 EJ** |

At full output the power fleet delivers about 31.6 EJ a year, so it repays its build energy in about 2 months and returns roughly 180 times that energy over 30 years. Even if manufacturing takes five times more energy than assumed, the return stays above 35 times.

### 13.4 Workforce

If labour is 30–40% of the build cost, SKIA employs roughly 510,000–680,000 people a year for 30 years, across solar cell and chip factories, launch sites, mission control, robotics and ground construction. Much of the ground work suits the skills of people who build and dismantle today's power plants and grids.

### 13.5 Space and programme risks

| Risk | Effect | Mitigation |
|---|---|---|
| Launch failures over thousands of flights | Lost payloads, debris in low orbit | Many small payloads, so no single flight matters much |
| Orbital debris and collisions | Damage to GEO and low-orbit satellites | Robotic debris removal, controlled disposal, short transit through low orbit |
| Solar storms and radiation | Solar cell wear, electronics faults | Shielding, redundancy, robots that swap worn panels |

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| Risk | Effect | Mitigation |
|---|---|---|
| Micrometeoroids | Punctured panels and film | Failure-tolerant design built from many small parts |
| Launch exhaust in the upper atmosphere | About 3,000 t of $CO_2$ per launch (approximate), plus water vapour and soot high up; ozone effects need study | Methane-oxygen fuel, later supply from the Moon |
| Hub or mirror failure | A rectenna loses half its supply, about 1.04 GW | Paired beams and storage (Section 11.9) |
| Sudden end of the shade | Rebound warming | Ramp-down limit of 0.1 W/m² a year (Section 12.1) |
| Political disputes | Delays or conflict over control | Treaty-based ownership and shared mission control |

### 13.6 Financials

| Cost item | Low | High | Assumption |
|---|---|---|---|
| Launch, about 43,600 flights | $0.22 trillion | $0.44 trillion | $5–10 million per flight |
| Power hardware, 2.38 million t | $1.2 trillion | $4.8 trillion | $500–2,000 per kg in mass production |
| Shade hardware, 1.98 million t | $0.2 trillion | $1.0 trillion | $100–500 per kg |
| Ground receivers, 481 sites | $1.0 trillion | $1.5 trillion | $2.0–3.1 billion each; NASA estimated $3.1 billion for one (NASA OTPS) |
| **Total build cost** | **$2.6 trillion** | **$7.7 trillion** | Over about 30 years |

Spread over 30 years that is $86–256 billion a year, 2.6–7.8% of the $3.3 trillion the world is set to invest in energy in 2025 (IEA). Operations and maintenance add roughly $50–150 billion a year.

The power fleet sells about 8,780 TWh a year. At $0.04–0.08 per kWh that earns $350–700 billion a year, or about $13 trillion over 30 years at $0.05. At $0.05, net income of about $290–390 billion a year repays the build cost within about 7–27 years of full operation. These figures leave out financing costs and the higher price of first-of-a-kind units.

If panels weigh 500 g/m² instead of 250 (Section 12.4), the space side grows to about 5.5 million t and 55,000 flights, and the total build cost rises to about $3.2–10.1 trillion. The power fleet's lifecycle emissions then rise to roughly 10 Gt, so it repays its carbon in about 2.6 years instead of under 2, and the net saving stays above 100 Gt over 30 years.

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### 13.7 Environmental return

| Carbon ledger, 30 years of operation | $CO_2$ (Gt) |
|---|---|
| Power fleet lifecycle emissions, including launches and manufacturing (26 g per kWh, from NASA's estimate for a similar system) | +6.9 |
| Shade fleet launches and manufacturing | +0.08 |
| Fossil electricity displaced at 0.45 kg per kWh | −118.5 |
| **Net** | **about −112** |

The net saving equals about three years of today's energy emissions, which reached 35.8 Gt in 2025 (Energy Institute, via DieselNet). Valued at the US EPA's social cost of carbon of $190 per tonne (Global Environmental Law Review), that avoided damage is worth about $21 trillion, several times the build cost. The fleet repays its own emissions in about 1.8 years.

The shade works differently. At today's 426 ppm of $CO_2$ (NOAA), 1 W/m² of shading cancels roughly the warming of 73 ppm, about 570 Gt of $CO_2$ in the air, for about 0.08 Gt emitted to build it. But it masks warming rather than removing $CO_2$, so oceans keep acidifying and the cooling lasts only while the fleet is maintained.

SKIA's own power can supply the missing piece: removing $CO_2$. Direct air capture needs about 4–10 GJ (1.1–2.8 MWh) per tonne, mostly as heat (Clean Energy Wire). Giving 10–20% of SKIA's output to capture plants removes about 0.3–1.6 Gt of $CO_2$ a year, or 9–48 Gt over 30 years.

### 13.8 Will SKIA outlast the damage of building it?

On its own ledger, yes, clearly. It repays its build energy in about 2 months and its carbon in under 2 years, and then gives about 28 years of net benefit: roughly 4 Gt of $CO_2$ avoided a year, about 11% of today's energy emissions, plus 0.8 °C of cooling at the cap.

Leaving Earth better than today takes all three SKIA pieces working together: the shade buys time, the power fleet cuts emissions, and capture plants powered by SKIA pull legacy $CO_2$ back out. Alone, SKIA cannot return the climate to its pre-industrial state; alongside the rest of the world's shift away from fossil fuels, it could be the difference between overshooting dangerous warming and recovering from it. The biggest uncertainties are cost, the upper-atmosphere effects of tens of thousands of launches, and whether nations can agree to share control.

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All derived figures (areas, masses, efficiencies, spot sizes, flight counts) were computed for this study from the equations in Section 3.

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